

EXAM 7 – SPRING 2014

24. (4 points)

The following information (\$000,000) is available for an insurer:

Earned premium	1,000
Expected losses	200
Acquisition expenses	150
Beginning surplus	2,000
Risk-free rate	3%
Investment rate of return	5%

The insurer is considering a reinsurance program for accident year 2015:

	<u>No Reinsurance</u>	<u>Reinsurance</u>
Reinsurance premium	0	450.0
Reinsurance commission	0	67.5
Expected recovery	0	110.0
Probability of distress	3.00%	0.40%

Assume the following:

- All premiums are collected at the beginning of the year.
- Losses are paid in full at the end of the year.
- All expenses are paid at the beginning of the year.
- There is no growth or change in the insured portfolio.
- The insurer cannot operate in a financially distressed state.
- Regulatory considerations are negligible.
- The firm value is the risk-adjusted present value of all future earnings.

a. (1.5 points)

Calculate the insurer's expected underwriting profit both with and without reinsurance.

b. (2.5 points)

Calculate the change in firm value of the insurer at the beginning of 2015 if the insurer purchases the reinsurance program.

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SAMPLE ANSWERS AND EXAMINER'S REPORT

QUESTION 24	
TOTAL POINT VALUE: 4	LEARNING OBJECTIVE: C6.b: Describe the rationale for, methods for, and effect of managing insurance and financial risks
SAMPLE ANSWERS	
Part a: 1.5 points	
<i>Sample Answer 1</i> w/o Reins: U/W Profit = EP – Loss – Expense = 1000 – 200 – 150 = 650 w/ Reins: U/W Profit = (1000 – 450) – (200 – 110) – (150 – 67.5) = 377.5 <i>Sample Answer 2</i> Without Reinsurance: U/W Profit = 1000 – 200 – 150 = 650 M With Reinsurance: U/W Profit = 650 – 450 + 67.5 + 110 = 377.5 M <i>Sample Answer 3</i> Gross underwriting profit = EP – loss – expense = 1,000 -200 – 150 = 650 Cost of Reinsurance = Reinsurance Prem – commission – recovery = 450 – 67.5 – 110 = 272.5 UW Profit with Reinsurance = 650 – 272.5 = 377.5	
Part b: 2.5 points	
<i>Sample Answer 1</i> Without: Inv Inc = [2000 + (1000 – 150)] × 0.05 = 142.5 D = (1-d)/(1+r) = (1-.03)/(1+.03) = 0.942 V = E × D/(1-D) = (650 + 142.5)(0.942/(1-0.942)) = 12,821 With: Inv Inc = (2000 + (1000 – 150 – 450 + 67.5)) × 0.05 = 123.4 D = (1 – 0.004)/1.03 = 0.967 V = (377.5 + 123.4)(0.967/(1 – 0.967)) = 14,678 Value increased 14,678 – 12,821 = 1,807 <i>Sample Answer 2</i> Without Reinsurance: E = 650 M + (2,000 + 1,000 – 150)(5%) = 792.5M D = (1-d)/ (1+r) = (1-3%)/(1+3%) = 0.97/1.03 , M = D/(1-D) = 16.167 V = E × M = (792.5M)(16.167) = 12,812.08M With Reinsurance: E = 377.5M + (2,000 + 1,000 – 450 + 67.5 – 150)(5%) = 500.875M	

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$$D = (1 - 0.4\%) / (1 + 3\%) = 0.996 / 1.03, M = D / (1 - D) = 29.294$$

$$V = E \times M = (500.875)(29.294) = 14,672.69M$$

$$\text{Change in Firm Value} = 14,672.69M - 12,812.08M = 1,860.61M$$

Sample Answer 3

No Reinsurance - U/W Profit = 650

Investible Assets = Capital + Prem – Expense = 2000 + 1000 – 150 = 2850

$$2850(0.05) = 142.5$$

Total Earnings = 792.5

Reinsurance U/W Profit = 377.5

Investible Assets = Capital + Prem – Expense + Commission = 550 - 150 + 67.5 + 2000 = 2467.5

$$2467.5(0.05) = 123.4$$

Total Earnings = 377.5 + 123.4 = 500.9

Risk Adj PV

No Reins

$$792.5(1 - 0.03) / (0.03 + 0.03) = 12,812$$

Reins

$$500.9(1 - 0.004) / (0.004 + 0.03) = 14,673$$

$$14,673 - 12,812 = 1,861$$

Sample Answer 4

	W/O Reinsurance	With Reinsurance
d =	3%	0.4%
D = (1 - d)/(1 + r)	0.97/1.03 = 0.9417	0.996/1.03 = 0.9670
K = (1 - D)/D	0.0619	0.0341
Inv Assets (incl premium and expense)	2000 + 1000 – 150	2000 + 550 – 82.5
Investment Income	(2000 + 1000 – 150) × 0.05 = 142.5	(2000 + 550 – 82.5) × 0.05 = 123.375
Earnings	650 + 142.5 = 792.5	377.5 + 123.375 = 500.875
V = E / k	792.5 / 0.0619 = 12,802.91	500.875 / 0.0341 = 14,688.42

$$\text{Change in firm value} = 14,688.42 - 12,802.91 = 1885.51 \text{ increase in value due to reinsurance}$$

Sample Answer 5

Investable assets

No Reinsurance = Surplus + EP – expense = 2000 + 1000 – 150 = 2,850

Reinsurance = 2850 – Reinsurance Prem + Commission = 2850 – 450 + 67.5 = 2,467.5

SAMPLE ANSWERS AND EXAMINER'S REPORT

	No Reinsurance	Reinsurance
UW Income	650	377.5
Investment Income = Inv Assets × 0.05	142.5	123.4
Earnings = UW Income + Inv Income	792.5	500.9
Prob distress d	3%	0.4%
$D = (1-d)/(1+r)$, $r=3\%$	0.942	0.967
$M = D/(1-D)$	16.17	29.29
$E \times M$	12,815	14,671

Change in Firm Value = 14,671 – 12,815 = 1,856

Sample Answer 6

$$D = (1 - d) / (1 + r)$$

W/O Reinsurance

$$D = (1 - 0.03)/(1 + 0.03) = 1 / (1+k) \rightarrow k = 0.0619$$

$$\text{Earnings} = \text{U/w Income} + \text{Investment Income} = 650 + (2000 + 1000 - 150) \times 0.05 = 792.5$$

$$BV = E \times (1+g)/(k-g), \quad g=0$$

$$E/k = 792.5/0.0619 = 12,803$$

W/ Reinsurance

$$D = (1 - 0.004)/(1 + 0.03) = 1/(1+k) \rightarrow k = 0.0341$$

$$\text{Earnings} = \text{U/W Income} + \text{Investment Income} = 377.5 + (2000 + 1000 - 450 + 67.5 - 150) \times 0.05 = 500.875$$

$$BV = E/k = 500.875/0.0341 = 14,688$$

$$\text{Chg BV} = 14,688 - 12,803 = 1,866$$

EXAMINER'S REPORT

Question 24 tested the candidates' ability to calculate and compare firm value under two scenarios, with and without reinsurance. The problem was set up to use the risk-adjusted present value of future earnings method as described in Venter and Underwood "Value of Risk Reduction".

In general, the candidates scored well on this question. Over half the candidates received full credit and most received nearly full credit.

Part a

Candidates were expected to calculate underwriting profit with and without reinsurance. In order to obtain full credit, the candidate had to show the correct underwriting profit with and without reinsurance as well as the components of premium, expense/commission, and losses for the reinsurance portion. 76% of candidates received full credit on this part.

Common errors included the following:

- Including investment income in the underwriting profit (about 10% of candidates)

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- Calculation errors
- Discounting losses

Part b

Candidates needed to compare firm value with and without reinsurance, which required that the same calculations be performed for each scenario. Candidates needed to understand all the components of the method. Full credit required that they accurately show calculations for invested assets, investment income, total earnings, the discount and perpetuity factors, calculation of firm value using those components, and the difference in firm value for the two scenarios. 69% of candidates received full credit.

Common errors included the following:

- Calculation errors
- Incorrect calculation of invested assets (omitting surplus or another component)
- Use of an incorrect investment rate or risk-free rate in a particular calculation